AP PRECALCULUS

Unit 1 • Worksheet 1.5C • Answer key

Worksheet 1.5C • Polynomial Functions and Complex Zeros

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

How many complex zeros, counted with multiplicity, does $p(x) = x^6 - 2x^3 + 5$ have?

(A) 3

(B) 5

(C) 6

(D) 2

Solution

Answer: (C). Degree 6.

Question 2

NO CALCULATOR

A polynomial with real coefficients has zeros 2 and $1 + 3i$. Which must also be a zero?

(A) -2

(B) $3 + i$

(C) $-1 - 3i$

(D) $1 - 3i$

Solution

Answer: (D). Conjugate pairs.

Question 3

NO CALCULATOR

What are the zeros of $p(x) = x^3 + 4x$?

(A) $0,\ 2,\ -2$

(B) $0,\ 2i,\ -2i$

(C) $0,\ 4i$

(D) 0 only

Solution

Answer: (B). $x(x^2 + 4)$.

Question 4

NO CALCULATOR

A polynomial of degree 5 with real coefficients has exactly two non-real zeros. How many real zeros (counted with multiplicity) does it have?

(A) 1
(B) 2
(C) 3
(D) 5

Solution
Answer: (C). $5 - 2 = 3$.

Question 5 (Free response)

NO CALCULATOR

Find all complex zeros: (a) $x^2 + 9$ (b) $x^4 - 16$ (c) $x^3 - x^2 + 4x - 4$ (hint: group).

Solution
(a) $\pm 3i$. (b) $(x^2 - 4)(x^2 + 4)$: $\pm 2, \pm 2i$. (c) $x^2(x - 1) + 4(x - 1) = (x - 1)(x^2 + 4)$: $1, \pm 2i$.